v. Reactive dyes:

Reactive dyes are suitable for mostly all fibres- cellulosic, wool, nylon, silk, acrylic, as well as blends. Bright colours with excellent wash fastness &colour fastness are obtained. Color fastness to crocking is excellent. **Example:** Cotton

Crocking: Crocking is tendency of a fabric or paint to give off the colour when the surface is rubbed.

4.7.2. Dyeing Methods

There are four steps involved by which colour may be applied to textile materials.

i. **Solutions:** Pigments or dye stuffs can be dispersed in the spinning solution where the fibres are manufactured. Solution dyed yarn is highly resistant to UV fade and shade changes. Solution dyed yarns are fully uniform in colour and typically do not vary from lot to lot. They are colour fast, resistant to multiple washing and mild bleaching.

ii. **Fibre dyeing:** Fibres are dyed in their loose state where the fibres can be less tangled and dyed thoroughly. Fibre dyeing is expensive to produce. There is better penetration of the dye into the fibre which produces a higher degree of colour fastness on fabrics. Direct, sulphur, vat and developed dyes are used on cellulose fibres.



Fibre dyeing

iii. **Yarn dyeing:** Yarns are dyed by three methods namely, skein dyeing, package dyeing and beam dyeing. Yarn dyeing is best way to adopt for large quantity dyeing.

iv. **Fabric dyeing:** Fabric dyeing consist of either piece dyeing the fabric or printing followed by an after treatment to fix the dye. Piece dyeing is nothing but most solid colour fabrics are dyed after the fabrics has been completed. Fabrics made of cellulosic fibres, cotton, rayon and flax are most frequently piece dyed.

4.8 Printing

Designs are applied on fabrics by means of printing. Printed fabrics are defined as those that have been decorated by a motif, pattern or design applied to the fabric after it has already been constructed(Marjory Joseph 1977). Printing can be done by two basic methods –Resist printing and direct printing.

4.8.1. Resist printing

Resist printing is done by preventing the dye to enter some specific portions of the fabric by some method.

i. **Tie and dye:** Fabrics are made into tiny puffs with some object inside and tied with a waxed thread wherever the dye has to be prevented. The fabric is immersed in the dye solution. If two or more colours are desired the thread is removed and the fabric re-tied. After drying the object is removed. Other methods of tie and dye includes holding the

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